

WATERTOWN, NY, USA

WMO: 726227

Lat: 43.989N

Lon: 76.026W

Elev: 97

StdP: 100.17

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94790

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.9	-20.8	-28.9	0.3	-24.4	-24.9	0.4	-20.2	12.7	-2.2	11.7	-1.6	1.9	80	0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	29.8	22.1	28.5	21.5	27.3	20.7	23.7	27.9	22.8	26.8	21.9	25.7	4.5	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.3	17.2	26.2	21.3	16.2	25.3	20.5	15.3	24.7	71.0	28.2	67.5	26.7	64.5	25.7	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.4	8.8	7.9	-33.0	32.3	3.7	1.5	-35.7	33.3	-37.8	34.2	-39.9	35.0	-42.5	36.0		
			-33.0	25.1	3.4	1.0	-35.4	25.8	-37.4	26.4	-39.4	26.9	-41.9	27.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.8	-6.6	-5.7	-0.6	6.4	13.0	18.0	20.8	20.0	16.0	9.8	3.6	-2.3
	DBStd	10.96	8.27	7.04	6.32	4.91	4.37	3.72	2.98	3.17	4.26	4.82	5.36	6.43
	HDD10.0	2089	515	439	333	130	18	0	0	0	4	64	203	383
	HDD18.3	4092	772	672	588	359	173	51	11	20	96	268	443	640
	CDD10.0	1273	1	1	3	22	112	239	334	309	183	58	9	1
	CDD18.3	234	0	0	0	1	9	41	87	71	24	2	0	0
	CDH23.3	1892	0	0	2	17	98	333	703	538	190	12	0	0
	CDH26.7	386	0	0	0	3	14	72	151	109	37	1	0	0
Wind	WSAvg	3.5	4.1	4.0	4.0	4.0	3.4	3.1	3.0	2.8	2.9	3.6	3.8	3.9
Precipitation	PrecAvg	815	62	48	60	62	71	67	54	80	86	72	88	71
	PrecMax	1078	189	139	145	135	143	158	120	196	188	154	182	170
	PrecMin	529	13	6	4	16	18	8	8	23	22	5	27	8
	PrecStd	140	42	29	30	34	34	35	30	36	38	32	40	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.6	13.5	20.2	25.7	28.5	30.9	31.5	31.3	30.1	24.9	19.8	15.9
		MCWB	12.8	8.9	14.2	16.2	20.2	22.6	23.6	22.6	22.2	18.0	14.2	12.8
	2%	DB	10.6	9.5	14.8	21.5	26.1	28.8	29.7	29.2	27.5	22.3	17.1	11.1
		MCWB	8.7	6.2	9.7	13.8	18.8	21.3	22.3	21.8	21.1	17.1	12.2	8.6
	5%	DB	6.9	6.6	11.9	18.0	23.9	27.0	28.4	27.8	25.5	20.0	14.6	8.8
		MCWB	4.8	3.9	8.0	11.4	17.3	20.2	21.8	21.0	19.9	15.6	11.1	6.6
	10%	DB	3.8	3.8	8.6	15.2	21.6	25.1	27.1	26.3	23.7	17.7	12.1	6.2
		MCWB	2.0	1.8	5.3	9.9	15.5	19.3	20.9	20.5	18.8	13.8	8.8	4.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.7	10.0	14.5	17.6	21.7	23.9	25.2	24.6	23.5	20.1	15.3	12.6
		MCDB	15.4	12.9	18.8	23.2	27.0	28.9	29.7	29.0	27.9	22.8	18.3	15.3
	2%	WB	8.6	6.7	10.7	14.9	20.0	22.5	23.9	23.2	22.0	17.9	13.3	9.0
		MCDB	10.4	9.1	13.8	19.9	24.7	27.1	28.1	27.2	26.1	21.4	16.2	10.4
	5%	WB	5.0	4.1	8.0	12.7	18.3	21.3	22.9	22.3	20.7	15.9	11.3	6.7
		MCDB	6.6	6.4	11.4	16.8	22.5	25.4	26.8	26.0	24.4	19.2	14.1	8.6
	10%	WB	2.2	2.0	5.5	10.4	16.6	20.2	22.1	21.4	19.6	14.4	9.1	4.2
		MCDB	3.7	3.6	8.3	14.6	20.5	23.9	25.7	24.9	22.9	17.5	11.7	6.1
Mean Daily Temperature Range	5% DB	MDBR	10.4	10.6	10.1	11.2	11.7	11.0	10.6	11.1	11.6	9.9	8.9	8.5
		MCDBR	10.4	11.5	13.9	16.4	14.5	12.9	11.8	12.4	12.6	12.1	12.2	10.2
		MCWBR	9.5	9.3	9.8	10.1	8.3	7.2	6.4	6.5	7.3	7.9	9.0	8.8
	5% WB	MCDBR	10.0	11.2	12.7	14.2	12.4	10.6	9.6	10.0	10.9	10.9	11.1	9.7
		MCWBR	9.4	9.3	9.4	9.7	7.9	6.4	6.0	6.1	6.8	7.7	8.8	8.8
Clear-Sky Solar Irradiance	taub		0.295	0.299	0.328	0.359	0.387	0.421	0.430	0.420	0.373	0.353	0.312	0.287
	taud		2.388	2.375	2.404	2.385	2.358	2.288	2.265	2.288	2.406	2.451	2.510	2.475
	Ebn at Noon		831	895	911	905	884	852	838	834	851	822	810	811
	Edh at Noon		76	92	103	114	120	130	131	124	102	86	68	64
All-Sky Solar Radiation	RadAvg		1.30	2.15	3.45	4.42	5.35	5.70	5.82	5.15	4.10	2.43	1.51	1.00
	RadStd		0.14	0.22	0.29	0.52	0.54	0.46	0.50	0.25	0.35	0.23	0.18	0.13

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page